

PROJECT BLUEPRINT

Premium Made-to-Measure Apparel Website

ASP.NET Core, Angular, Bootstrap ও PostgreSQL-ভিত্তিক আন্তর্জাতিক
luxury tailoring platform-এর পূর্ণাঙ্গ পরিকল্পনা

Backend: **api** • Frontend: **ui**

প্রস্তুতের তারিখ: ২০ সেপ্টেম্বর ২০২৬

১. প্রকল্পের লক্ষ্য

বাংলাদেশে তৈরি premium made-to-measure suit, shirt, trouser ও women’s dress US/EU গ্রাহকের কাছে বিক্রির জন্য একটি আধুনিক D2C platform। সাইটটি luxury craftsmanship, personalized fitting, secure payment, আন্তর্জাতিক delivery এবং premium after-sales support তুলে ধরবে।

মূল business flow: Product নির্বাচন → customization → measurement → checkout → measurement review → tailoring/quality control → shipment → alteration support।

২. Technology stack

অংশ	প্রস্তাবিত প্রযুক্তি
Backend	ASP.NET Core Web API, Clean Architecture, Entity Framework Core, ASP.NET Core Identity, FluentValidation, Serilog, Swagger/OpenAPI
Frontend	Angular standalone components, Bootstrap 5, SCSS, Reactive Forms, Signals, route guards, HTTP interceptor, lazy loading, Angular SSR
Database	PostgreSQL; UUID primary key, UTC timestamps, monetary value numeric(18,2), indexes ও concurrency control
Background/Realtime	Hangfire jobs; প্রয়োজনে SignalR notification
Storage	AWS S3/Cloudinary/S3-compatible storage; private files-এর signed URL
Testing	xUnit unit/integration tests; Angular component ও end-to-end tests
Deployment	Docker, Nginx, managed PostgreSQL, staging/production environments, CI/CD

৩. Repository structure

```
root/
├─ api/
│  └─ src/
│     ├── TailorShop.Api/
│     ├── TailorShop.Application/
│     ├── TailorShop.Domain/
│     └─ TailorShop.Infrastructure/
│  └─ tests/
│     └─ Dockerfile
├─ ui/
│  └─ src/app/{core,shared,layouts,features,admin}/
│     └─ Dockerfile
├─ deployment/{nginx,docker-compose.yml}
├─ docs/
└─ README.md
```

8. User roles ও permission

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Roles: Customer, Admin, SuperAdmin, Tailor, OrderManager, ContentManager ও SupportAgent। শুধু role check নয়, granular permission ব্যবহার করতে হবে: products.manage, orders.manage, measurements.review, payments.view, refunds.manage, content.manage, reports.view, users.manage এবং settings.manage।

৫. Authorization model

- Default/fallback policy: authenticated user আবশ্যিক।
- Homepage, catalog ও product detail-এর read-only endpoint public থাকবে; rate limiting এবং caching থাকবে।
- Cart, wishlist, measurements, checkout, orders ও account endpoint Customer authorization চাইবে।
- Admin endpoint role + permission policy দিয়ে সুরক্ষিত হবে।
- Payment/shipping webhook JWT ব্যবহার করবে না; provider signature এবং idempotency যাচাই করবে।

কেন কিছু public endpoint দরকার: সব endpoint login-required হলে নতুন visitor product বা homepage দেখতে পারবে না। তাই public read-only endpoint ছাড়া সব API secured রাখা হবে।

৬. Customer website modules

Storefront ও content

- Luxury homepage, Men/Women collections, category, search, filters ও sorting
- Product details, fabric provenance, craftsmanship, delivery estimate এবং related products
- How It Works, Craftsmanship, Fabric Story, Lookbook/Journal, FAQ ও policy pages
- Country/currency selector, SEO metadata, sitemap, newsletter ও cookie consent

Product customization

Fabric, color/pattern, fit, jacket/lapel/button/vent/lining, trouser style, shirt collar/cuff, dress neckline/sleeve/length, monogram, personal notes, rush production ও luxury packaging। প্রতিটি option price modifier সমর্থন করবে এবং order-এর সঙ্গে immutable snapshot থাকবে।

Measurement system

- Guided manual measurement wizard, inches/cm, instructional media ও expected-range validation
- Self-measure, local tailor এবং virtual fitting—তিনটি পথ
- Front/side photos, fit preference, saved profiles, version history
- Admin/tailor approval ও correction request workflow
- পরবর্তী release-এ common provider interface-এর মাধ্যমে AI sizing integration

Customer account

Profile, addresses, measurement profiles, wishlist, cart, checkout, order history, production timeline, tracking, invoice, virtual consultation, support conversation, alteration-credit claim ও review।

৭. Order ও production workflow

Cart → Checkout → Payment → Measurement Submitted → Measurement Review
 → Fabric Allocated → Pattern → Cutting → Tailoring → First QC
 → Finishing → Final QC → Premium Packaging → Shipped → Delivered
 → Alteration Support / Completed

প্রতিটি stage-এ customer-visible status, internal status, timestamp, responsible staff, optional photo, staff note ও notification থাকবে।

৮. Admin dashboard

- **Overview:** sales, revenue, AOV, new customers, pending measurements, delayed orders, low-stock fabrics এবং upcoming fittings
- **Catalog:** products, categories, variants, fabrics/batches, collections, images, options, price modifiers ও SEO
- **Orders:** payment, customization, measurements, production, QC, shipment, invoice, cancellation/refund
- **Production board:** stage-based Kanban, staff assignment, deadlines ও delay warning
- **Measurement review:** range warning, photos, notes, approve/correction/fitting actions
- **Operations:** customers, staff/permissions, coupons, consultations, claims, reviews, CMS, reports, settings ও audit logs

৯. প্রধান database entities

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Domain	Entities
Identity	ApplicationUser, Role, Permission, RolePermission, RefreshToken, Address
Catalog	Category, Product, ProductVariant, ProductImage, Collection, Fabric, FabricBatch, CustomizationGroup, CustomizationOption, PriceModifier
Shopping	Cart, CartItem, CartItemCustomization, WishlistItem, Coupon, CouponUsage, GiftCard
Measurements	MeasurementProfile, MeasurementEntry, MeasurementPhoto, MeasurementReview, MeasurementVersion
Orders	Order, OrderItem, OrderItemCustomization, OrderAddress, OrderStatusHistory, ProductionTask, QualityCheck, Payment, Refund, Shipment
Support/Content	ConsultationBooking, SupportTicket, SupportMessage, AlterationClaim, Review, Notification, CmsPage, Banner, MediaAsset
System	TaxRule, CurrencyRate, EmailTemplate, SystemSetting, AuditLog, WebhookEvent

50. API endpoint blueprint

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Authentication

POST /api/v1/auth/register	POST /api/v1/auth/login
POST /api/v1/auth/refresh	POST /api/v1/auth/logout
POST /api/v1/auth/verify-email	POST /api/v1/auth/forgot-password
POST /api/v1/auth/reset-password	GET /api/v1/auth/me

Public catalog

GET /api/v1/storefront/home	GET /api/v1/categories
GET /api/v1/collections	GET /api/v1/collections/{slug}
GET /api/v1/products	GET /api/v1/products/{slug}
GET /api/v1/fabrics	GET /api/v1/pages/{slug}

Customer

GET/PUT	/api/v1/account/profile
CRUD	/api/v1/account/addresses
GET/POST/PUT/DELETE	/api/v1/cart/items
GET/POST/PUT	/api/v1/measurements
POST	/api/v1/measurements/{id}/submit
POST	/api/v1/measurements/{id}/photos
POST	/api/v1/checkout/preview
POST	/api/v1/checkout/create-payment
POST/GET	/api/v1/orders
GET	/api/v1/orders/{id}
POST/GET	/api/v1/consultations
POST	/api/v1/alteration-claims

Admin

GET	/api/v1/admin/dashboard
CRUD	/api/v1/admin/{products,categories,collections,fabrics}
CRUD	/api/v1/admin/{customization-groups,customization-options}
GET	/api/v1/admin/orders
POST	/api/v1/admin/orders/{id}/{status,assign,quality-check,shipment}
GET	/api/v1/admin/measurements/pending
POST	/api/v1/admin/measurements/{id}/{approve,request-correction}
CRUD	/api/v1/admin/{users,roles,coupons,pages,banners}
GET	/api/v1/admin/reports/sales
GET	/api/v1/admin/audit-logs
GET/PUT	/api/v1/admin/settings

Webhooks

POST	/api/v1/webhooks/payments/{provider}
POST	/api/v1/webhooks/shipping/{provider}
POST	/api/v1/webhooks/sizing/{provider}

১১. Angular structure

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```
app/
├─ core/{auth,guards,interceptors,services,models}
├─ shared/{components,directives,pipes,validators}
├─ layouts/{storefront-layout,account-layout,admin-layout}
├─ features/{home,catalog,product-details,product-builder,card,
            checkout,measurements,account,orders,consultations,content}
└─ admin/{dashboard,catalog,orders,measurements,customers,cms,reports,settings}
```

১২. Security checklist

- Short-lived access token, rotating refresh token ও HttpOnly/Secure cookie
- Email verification, strong password, logout, rate limiting এবং admin 2FA
- CORS allow-list, HTTPS, security headers, DTO validation ও safe error responses
- Upload-এর MIME/signature/size validation; private storage এবং signed URL
- Payment card data database-এ নয়; provider-hosted/tokenized payment flow
- Webhook signature, idempotency key ও duplicate-event protection
- Audit logs, encrypted secrets, sensitive-data-safe logging
- Automated backup, restore test, dependency/security scan

১৩. Development phases

Phase 1 — Foundation: requirements, repository, API/UI projects, Docker, PostgreSQL, authentication, permissions, logging, error handling ও CI.

Phase 2 — Catalog & CMS: categories, products, fabrics, customizations, collections, media, storefront, SEO ও CMS.

Phase 3 — Measurements: guided wizard, profiles, photos, validation, admin review, correction ও virtual fitting.

Phase 4 — Commerce: cart, coupons, address, price calculation, payment abstraction, checkout, orders, email ও invoice.

Phase 5 — Production & Delivery: production stages, assignments, QC, packaging, shipment, tracking ও notifications.

Phase 6 — Premium Support: alteration claims, support tickets, reviews, wishlist ও consultation calendar.

Phase 7 — Hardening: analytics, reports, accessibility, performance, security এবং automated tests.

Phase 8 — Launch: production infrastructure, domain/SSL, live integrations, monitoring, backups, UAT ও release.

MVP: Phase 1–5। AI sizing, advanced recommendation, loyalty এবং sophisticated multi-currency settlement পরবর্তী release-এ রাখা যাবে।

১৪. Testing ও quality gates

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- Unit tests: pricing, authorization, measurement validation ও status transitions
- Integration tests: auth, catalog, checkout, payment webhook ও order lifecycle
- E2E tests: browse → customize → measurement → checkout → order tracking
- Permission tests: প্রত্যেক admin role কেবল অনুমোদিত action করতে পারে
- Responsive, accessibility, cross-browser, load ও security testing
- Staging UAT approve না হলে production release নয়

১৫. Launch checklist

- Brand assets, product photography, copy, size/measurement instructions চূড়ান্ত
- Domain, SSL, production database, storage, email এবং DNS প্রস্তুত
- Payment, tax/duty messaging, courier ও tracking live verification
- Privacy, terms, shipping, returns ও alteration policy legal review
- Backups, alerts, error monitoring, analytics ও cookie consent সক্রিয়
- Admin training, order runbook, refund/claim process এবং incident contacts
- একটি সম্পূর্ণ test order দিয়ে payment থেকে delivery status পর্যন্ত rehearsal

১৬. Recommended delivery order

1. Scope ও data model freeze
2. Authentication/authorization foundation
3. Admin catalog → public storefront
4. Product configurator → measurements
5. Cart/checkout/payment → order lifecycle
6. Production/QC/shipping → customer tracking
7. Support/CMS/reports → hardening → launch

এই document-এ design-generation prompt অন্তর্ভুক্ত করা হয়নি। এটি শুধুমাত্র product, engineering, security, operations ও delivery plan।